

Khushi Raj

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Profile

I am a fourth-year undergraduate pursuing a BS–MS degree in Physics at IISER Berhampur, with a strong research interest in Quantum Information Theory and Quantum foundations.

Education

Indian Institute of Science Education and Research Berhampur *BS-MS* *Sept 2023 – Present*

- **Coursework and Labs in Physics:** Quantum Mechanics I, Quantum Mechanics II, Mathematical Methods I, Mathematical Methods II, Statistical Mechanics, Thermal Physics, Electromagnetism, Classical Mechanics, Waves and Optics, Optics Lab, Basic Electronics Lab.
- **Additional Coursework:** Mathematics, Computer and Data Science, Chemistry, Biology, Earth and Environmental Science, and Humanities and Social Science.
- CPI: 8.0/10

Internship

Prof. Ritabrata Sengupta (IISER Berhampur) *May 2026 – Present*

- Studied the theory of Mutually Unbiased Bases (MUBs) through the construction based on Weyl operators and maximal commuting classes of unitary operators, following the approach of Bandyopadhyay *et al.*
- Constructed complete sets of MUBs in prime and prime-power dimensions, and analyzed why this construction does not extend directly to composite dimensions.
- Investigated the operator-algebraic framework for MUBs presented by K. R. Parthasarathy in *On Estimating the State of a Finite Level Quantum System*, including the concepts of weakly and strongly mutually unbiased bases.
- Constructed and analyzed examples of weakly and strongly mutually unbiased bases, and studied Parthasarathy's characterization of MUBs using maximal abelian subspaces of matrix algebras.
- Continuing the project by studying further developments in the theory of mutually unbiased bases, with a prospective focus on constructions over p -adic fields.

Prof. Utpal Roy (IIT Patna) *Jun 2025 – July 2025*

- Studied the foundational concepts of quantum computation by completing the first four chapters of *Quantum Computation and Quantum Information* by Nielsen and Chuang.
- Learned Qiskit and implemented various quantum protocols to gain hands-on experience with quantum programming.
- Explored quantum machine learning, focusing on amplitude encoding techniques.
- Conducted an in-depth study of research papers on Quantum Key Distribution (QKD).
- Presented key findings and insights from the literature through multiple technical presentations.

Skills

Programming Languages: Python, MATLAB, JavaScript, Qiskit

Tools and Technologies: Git and Github, Jupyter, SQL, MongoDB, PHP